

Wei-Kai Li Academic CV

Professor / Doctoral Supervisor / Taishan Scholars Young Expert

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School of Computer Science and Artificial Intelligence, Shandong Jianzhu University / School of Mathematics and Statistics, Chongqing Jiaotong University / Panoramic Medical Imaging Center

Education

1. 2018–2022 / PhD / Computer Science and Technology / Nanjing University of Aeronautics and Astronautics
2. 2015–2018 / Master's / Computer Technology and Applications / Chongqing Jiaotong University
3. 2011–2015 / Bachelor's / Information and Computing Science / Chongqing Jiaotong University

Research Projects

1. Unsupervised Domain Adaptation beyond Domain Alignment and Its Extensions / NSFC Young Scientists Fund / Principal Investigator
2. Advanced School in Applied Machine Learning / NSFC International Cooperation and Exchange / Principal Investigator
3. Multimodal Neurophysiological Mechanisms of Heterogeneous Brain Networks in Depression and Digital-Twin Closed-Loop TMS Robotics / Shandong Major Basic Research Project / Principal Investigator
4. Exploration of Unsupervised Domain Adaptation beyond Domain Alignment / MIIT Key Laboratory Open Project / Principal Investigator
5. Early Identification of Parkinson's Disease Using PET/MR Multimodal Brain Networks / Hubei Key Laboratory of Molecular Imaging / Principal Investigator
6. Multiscale Analysis of Autism Using Multimodal Imaging / Chongqing Education Commission Young Research Fund / Principal Investigator
7. Brain Network Disease Diagnosis Based on Multicenter Transfer Learning / Chongqing Natural Science Foundation General Project / Principal Investigator
8. Probabilistic Synchronization and Secure Control of Stochastic Complex Networks / Chongqing Natural Science Foundation Joint Fund / Researcher
9. Investigating and Extending the Effectiveness of Self-Supervised Learning / NSFC General Program / Researcher

Publications

1. 2026 / Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View. H Qin, G Hong, D Ma, Z Wang, L Zhao, W Li, X Xu. CNS Neuroscience & Therapeutics 32 (9), e71158, 2026
2. 2026 / Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI. Z Zheng, X Chen, H Jiang, R Tian, Q Cai, J Liu, X Chen, W Li, Y Wang. arXiv preprint arXiv:2607.29531, 2026
3. 2026 / Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction. J Yang, W Li, K Qian, Z Cheng. Neurocomputing, 134626, 2026
4. 2026 / Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation. L Cui, Y Wu, Y Tang, W Li. IEEE Transactions on Computational Social Systems, 2026
5. 2026 / Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis. D Ma, L Peng, L Zhang, W Li, X Gao. Frontiers in Psychiatry 17, 1846996, 2026

6. 2026 / TumorAL: Evidence-aware active learning for 3D tumor segmentation. H Wang, J Leng, S Li, Y Zhao, W Li, W Li, X Gao. *Neurocomputing*, 134239, 2026
7. 2026 / Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns. Y Ruan, L Guan, L Peng, L Zhao, Y Huang, W Li, X Gao. *CNS Neuroscience & Therapeutics* 32 (6), e70951, 2026
8. 2026 / Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease. Q Wang, W Li, J Wan, L Wei, Y Xia, Y Hua, K Zhou, D Qie, W Li, Y Li. *Journal of Cardiovascular Development and Disease* 13 (6), 228, 2026
9. 2026 / Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq. X Chen, Q Zou, Q Cai, X Chen, W Li, Y Wang. *PLOS Computational Biology* 22 (5), e1014223, 2026
10. 2026 / Bridging the modality reliability gap in drug-target interaction prediction via a confidence-aware multimodal fusion framework. J Yang, J Zhang, K Qian, Q Yang, W Li, Z Cheng. *Proceedings of the AAAI Conference on Artificial Intelligence* 40 (32), 27529 ..., 2026
11. 2026 / Test-time few-shot object detection via dynamic prototype fusion. Y Wu, Y Li, H Wei, W Li, Y Tang. *IEEE Transactions on Cybernetics*, 2026
12. 2026 / Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect. Q Wang, Y Xia, L Wei, P Yang, K Zhou, Y Hua, W Li, Y Li. *Cardiology in the Young*, 1-11, 2026
13. 2025 / Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation. H Ma, HS Gan, H Mi, P Xu, Y Liu, W Li. *2025 IEEE International Conference on Big Data (BigData)*, 7503-7510, 2025
14. 2025 / More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease. D Li, J Yao, W Li, Z Guo, X Lv, L Chen, J Shi, YC Chen, J Ye. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 111578, 2025
15. 2025 / Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis. H Song, H Qin, W Li. *2025 International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
16. 2025 / MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning. Y Lu, N Tian, W Li. *2025 International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
17. 2025 / Low-light image enhancement using dual cross attention. Y Ruan, H Ma, D Ma, W Li, X Wang. *Engineering Applications of Artificial Intelligence* 159, 111501, 2025
18. 2025 / Auto-weighted projective one-step multi-view clustering. X Mou, W Li, B Li, J Hu. *Neurocomputing*, 131960, 2025
19. 2025 / Revisiting Multi-Modal Alignment: In Distribution View. W Li, N Tian, Y Li, Y Tang. *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC ...)*, 2025
20. 2025 / Quantitative susceptibility mapping shows alterations of brain iron content in children with autism spectrum disorder: a whole-brain analysis. X Xu, Y Li, H Lan, N Ding, W Li, G Zheng, X Song. *BMC psychiatry* 25 (1), 826, 2025
21. 2025 / Cross-modal medical image generation from MRI to PET using robust generative adversarial network. Y Yang, B Li, W Cao, X Chen, W Li. *Expert Systems with Applications*, 129287, 2025
22. 2025 / Leveraging multi-source and heterogeneous signals for fatigue detection. L Cui, Y Wu, T Ying, W Li. *arXiv e-prints*, arXiv: 2507.16859, 2025
23. 2025 / PSMA PET/CT based multimodal deep learning model for accurate prediction of pelvic lymph-node metastases in prostate cancer patients identified as candidates for extended Q Ma, B Chen, R Seifert, R Zhou, L Xiao, J Yang, A Rominger, K Shi, W Li, *European Journal of Nuclear Medicine and Molecular Imaging* 52 (6), 2063-2075, 2025
24. 2025 / Timestamp calibration for time-series single cell rna-seq expression data. X Chen, S Lin, X Chen, W Li, Y Li. *Journal of Molecular Biology* 437 (9), 169021, 2025
25. 2025 / Iterative neural networks for improving memory capacity. X Chen, D Lin, Z Li, W Li. *Neural Networks* 182, 106936, 2025
26. 2024 / The hidden Markov model reveals the changes in brain dynamics among patients with end-stage renal disease under different dialysis methods. D Sun, W Li, H Yu, X Chen, C Zhang. *Journal of Cyber-Physical-Social Intelligence* 3 (1), 10-10, 2024

27. 2024 / An Improved Algorithm for Spiking Neural Networks with Multi-Scale Attention Coding. S Chen, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
28. 2024 / Improved domain-adversarial neural networks with mixup and pseudo-labeling. X Xiao, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
29. 2024 / Dynamic Graph Contrastive Learning for Neuro Disease Identification using Resting-State fMRI. Y Jiang, W Li, X Chen. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
30. 2024 / Synthesizing A β -PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer's disease. Y Yang, B Li, W Li, H Chen, W Cao. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
31. 2024 / Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification. H Qin, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
32. 2024 / Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization. D Sun, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
33. 2024 / Multi-view comparative self-supervised graph learning for neuro-disease diagnosis using functional brain networks. N Tian, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
34. 2024 / Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification. J Deng, L Guan, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
35. 2024 / Efficient and accurate capsule networks with b-spline-based activation functions. L Mou, X Xiao, W Cao, W Li, X Chen. 2024 International Conference on New Trends in Computational Intelligence ..., 2024
36. 2024 / The characteristic patterns of individual brain susceptibility networks underlie Alzheimer's disease and white matter hyperintensity-related cognitive impairment. H Chen, J Xu, W Li, Z Hu, Z Ke, R Qin, Y Xu. *Translational Psychiatry* 14 (1), 177, 2024
37. 2024 / TIDE: Test-time few-shot object detection. W Li, H Wei, Y Wu, J Yang, Y Ruan, Y Li, Y Tang. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 54 (11), 6500-6509, 2024
38. 2024 / Multipattern graph convolutional network-based autism spectrum disorder identification. W Zhou, M Sun, X Xu, Y Ruan, C Sun, W Li, X Gao. *Cerebral Cortex* 34 (3), bhae064, 2024
39. 2024 / Limbic/paralimbic connection weakening in preschool autism-spectrum disorder based on diffusion basis spectrum imaging. T Yi, W Li, W Wei, G Wu, G Jiang, X Gao, K Jin. *The European journal of neuroscience*, 2024
40. 2024 / Robust support vector machine based on sample screening. J Guo, W Li, J Hu. *The 14th International Conference on Information Science and Technology*, 2024
41. 2024 / Altered intra-and inter-network connectivity in autism spectrum disorder. R Zhou, C Sun, M Sun, Y Ruan, W Li, X Gao. *Aging* 16, 2024
42. 2024 / Machine learning models using symptoms and clinical variables to predict coronary artery disease on coronary angiography. Y Yu, W Li, J Wu, X Hua, B Jin, H Shi, Q Chen, J Pan. *Advances in Interventional Cardiology/Postępy w Kardiologii Interwencyjnej* ..., 2024
43. 2024 / Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data. J Yang, X Xu, M Sun, Y Ruan, C Sun, W Li, X Gao. *Cerebral Cortex* 34 (1), bhad477, 2024
44. 2024 / Effects of individual metabolic brain network changes co-affected by T2DM and aging on the probabilities of T2DM: protective and risk factors. YL Li, JJ Wu, WK Li, X Gao, D Wei, X Xue, XY Hua, MX Zheng, JG Xu. *Cerebral Cortex* 34 (1), bhad439, 2024
45. 2024 / Topological properties analysis and identification of mild cognitive impairment based on individual morphological brain network connectome. X Xu, P Chen, W Li, Y Xiang, Z Xie, Q Yu, Y Tang, P Wang. *Cerebral Cortex* 34 (1), bhad450, 2024
46. 2023 / Wavelet transform-based frequency self-adaptive model for functional brain network. Y Ding, X Xu, L Peng, L Zhang, W Li, W Cao, X Gao. *Cerebral Cortex* 33 (22), 11181-11194, 2023
47. 2023 / PageRank algorithm based on dynamic damping factor. Z HaoLin, H Jin, L WeiKai. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ..., 2023

48. 2023 / An improved AdaBoost method in imbalanced data Learning. T Li, X Chen, W Li. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ..., 2023
49. 2023 / Minimum error entropy high-order extended Kalman filter. Z Cheng, X Chen, H Li, W Li, D Lin. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ..., 2023
50. 2023 / New insights into the disorder of brain connectivity in schizophrenia. W Li, MM Al-Amin, A Nair. *Frontiers in Neuroimaging* 2, 1266695, 2023
51. 2023 / The reconfiguration pattern of individual brain metabolic connectome for Parkinson's disease identification. W Li, Y Tang, L Peng, Z Wang, S Hu, X Gao. *MedComm* 4 (4), e305, 2023
52. 2023 / TIToK: A solution for bi-imbalanced unsupervised domain adaptation. Y Wang, Q Chen, Y Liu, W Li, S Chen. *Neural Networks* 164, 81-90, 2023
53. 2023 / Clinical application of machine learning methods in psychiatric disorders. X Liu, Z Xu, W Li, Z Zhou, X Jia. *Frontiers in psychiatry* 14, 1209615, 2023
54. 2023 / Functional and structural brain network construction, representation and application. W Li, Z Wang, S Hu, C Chen, M Liu. *Frontiers in neuroscience* 17, 2023
55. 2022 / Unsupervised domain adaptation with progressive adaptation of subspaces. W Li, S Chen. *Pattern Recognition* 132, 108918, 2022
56. 2022 / Jacobian norm for unsupervised source-free domain adaptation. W Li, M Cao, S Chen. *arXiv preprint arXiv:2204.03467*, 2022
57. 2022 / Machine learning-derived multimodal neuroimaging of presurgical target area to predict individual's seizure outcomes after epilepsy surgery. Y Tang, W Li, L Tao, J Li, T Long, Y Li, D Chen, S Hu. *Frontiers in Cell and Developmental Biology* 9, 669795, 2022
58. 2022 / Human-Guided Functional Connectivity Network Estimation for Chronic Tinnitus Identification: A Modularity View. WK Li, YC Chen, XW Xu, X Wang, X Gao. *IEEE journal of biomedical and health informatics*, 2022
59. 2022 / Partial Domain Adaptation without Domain Alignment. W Li, S Chen. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022
60. 2022 / Machine learning based on the multimodal connectome can predict the preclinical stage of Alzheimer's disease: a preliminary study. H Chen, W Li, X Sheng, Q Ye, H Zhao, Y Xu, F Bai, *European Radiology* 32 (1), 448-459, 2022
61. 2021 / Multiple connection pattern combination from single-mode data for mild cognitive impairment identification. W Li, X Xu, Z Wang, L Peng, P Wang, X Gao. *Frontiers in Cell and Developmental Biology* 9, 782727, 2021
62. 2021 / Morphological, structural, and functional networks highlight the role of the cortical-subcortical circuit in individuals with subjective cognitive decline. X Xu, T Wang, W Li, H Li, B Xu, M Zhang, L Yue, P Wang, S Xiao. *Frontiers in Aging Neuroscience* 13, 688113, 2021
63. 2021 / Characteristics of multimodal brain connectomics in patients with schizophrenia and the unaffected first-degree relatives. X Lin, WK Li, G Dong, Q Wang, H Sun, J Shi, Y Fan, P Li, L Lu. *Frontiers in cell and developmental biology* 9, 631864, 2021
64. 2020 / Leave zero out: Towards a no-cross-validation approach for model selection. W Li, C Geng, S Chen. *arXiv preprint arXiv:2012.13309*, 2020
65. 2020 / Effective and accurate diagnosis of subjective cognitive decline based on functional connection and graph theory view. X Xu, W Li, M Tao, Z Xie, X Gao, L Yue, P Wang. *Frontiers in neuroscience* 14, 577887, 2020
66. 2020 / Functional connectivity network estimation with an inter-similarity prior for mild cognitive impairment classification. W Li, X Xu, W Jiang, P Wang, X Gao. *Aging (Albany NY)* 12 (17), 17328, 2020
67. 2020 / The effect of jet lag on the human brain: A neuroimaging study. F Zhang, W Li, H Li, S Gao, JA Sweeney, Z Jia, Q Gong. *Human brain mapping* 41 (9), 2281-2291, 2020
68. 2020 / Group similarity constraint functional brain network estimation for mild cognitive impairment classification. X Gao, X Xu, X Hua, P Wang, W Li, R Li. *Frontiers in Neuroscience* 14, 165, 2020

69. 2020 / Feature selection and combination in the functional brain connectome for discrimination of mild cognitive impairment and analyses of altered brain patterns. X Xu, W Li, J Mei, M Tao, X Wang, Q Zhao, X Liang, W Wu, D Ding, Frontiers in aging neuroscience 12, 28, 2020
70. 2020 / ■■■■■■■■■■■■■■■■■■■■. ■■■■ ■■■■ ■■■■ ■■■■ ■■■■. ■■■■■■■■■■ 2 (1), 80-87, 2020
71. 2019 / Toward a better estimation of functional brain network for mild cognitive impairment identification: a transfer learning view. W Li, L Zhang, L Qiao, D Shen. IEEE journal of biomedical and health informatics 24 (4), 1160-1168, 2019
72. 2019 / The altered reconfiguration pattern of brain modular architecture regulates cognitive function in cerebral small vessel disease. R Liu, H Chen, R Qin, Y Gu, X Chen, J Zou, YC Jiang, W Li, F Bai, Frontiers in neurology 10, 324, 2019
73. 2019 / Functional brain network estimation with time series self-scrubbing. W Li, L Qiao, L Zhang, Z Wang, D Shen. IEEE journal of biomedical and health informatics 23 (6), 2494-2504, 2019
74. 2019 / ■■■■■■■■■■■■■■■■■■■■. ■■■■ ■■■■ ■■■■ ■■■■. ■■ 2, 2019
75. 2018 / ■■■■■■■■■■■■■■■■■■■■.. ■■■■ ■■■■ ■■. Computer Engineering & Science/Jisuanji Gongcheng yu Kexue 40 (4), 699, 2018
76. 2018 / Simultaneous estimation of low-and high-order functional connectivity for identifying mild cognitive impairment. Y Zhou, L Qiao, W Li, L Zhang, D Shen. Frontiers in neuroinformatics 12, 337293, 2018
77. 2017 / Remodeling Pearson's correlation for functional brain network estimation and autism spectrum disorder identification. W Li, Z Wang, L Zhang, L Qiao, D Shen. Frontiers in neuroinformatics 11, 55, 2017
78. Year not listed / Kola: Carefully benchmarking world knowledge of large language models, 2024. J Yu, X Wang, S Tu, S Cao, D Zhang-Li, X Lv, H Peng, Z Yao, X Zhang, URL <https://arxiv.org/abs/2306.09296>
79. Year not listed / An Improved Efficient Algorithm for Spiking Neural Network Through Dataset Distillation. S Chen, X Chen, W Cao, W Li. Available at SSRN 4836784
80. Year not listed / Graph Adversarial Discriminative Domain Adaptation with Category-Level Alignment. X Xiao, X Chen, W Li, Z Li, W Cao, S Zhang. Available at SSRN 4898772

Academic Service

1. Area Chair, IEEE SMC 2026
2. Area Chair, IFAC CPHS 2020
3. Area Chair, ICCSI 2022–2025
4. Convener, Embodied Intelligence Conference Parallel Forum
5. Youth Editorial Board, Journal of Data Acquisition and Processing
6. Youth Editorial Board, Meta-Radiology
7. Guest Editor, Frontiers in Neuroscience and related journals
8. Outstanding Reviewer, Journal of Data Acquisition and Processing
9. Youth Editorial Board, Aging & Disease
10. Youth Editorial Board, Brain-X
11. Youth Editorial Board, Cog
12. Corresponding Member, CAAI Machine Learning Committee
13. Deputy Secretary-General, CICC Embodied Intelligence Committee
14. Member, CAA Human–Machine Education Committee
15. Member, CSS Virtual Technology and Applications Committee
16. Committee Member, Youth Workshop on Medical Imaging
17. Council Member, Chongqing Mathematical Society

Personal Recognition

1. 2026 / CPSI Best Paper Award
2. 2025 / Taishan Scholars Young Expert
3. 2025 / CPSI Best Paper Award
4. 2025 / Outstanding Faculty Advisor, China Robot and AI Competition
5. 2023 / ICCSI Best Paper Finalist Award
6. 2022 / National Scholarship for Doctoral Students

Student Awards

1. 2026 / Jingxuan Cui and Coauthors: CPSI 2026 Best Paper Award
2. 2025 / First Prize, China Robot and AI Competition National Finals
3. 2025 / First Prize, China Robot and AI Competition Chongqing Selection
4. 2025 / National Third Prize, National Digital Art Design Competition
5. 2025 / MCM/ICM Honorable Mention and Meritorious Winner
6. 2025 / First Prize, China International College Students' Innovation Competition: Paying Xianzhi
7. 2025 / First Prize, China International College Students' Innovation Competition: Yingluo Zhizhen
8. 2025 / Second Prize, iCAN Campus Competition
9. 2025 / Third Prize, China Graduate Electronics Design Competition, Southwest Region
10. 2024 / MCM/ICM Honorable Mention and Meritorious Winner
11. 2024 / First and Third Prizes, China International College Students' Innovation Competition, Campus Round
12. 2024 / Second Prize, China Robot and AI Competition National Finals
13. 2024 / Second Prize, Chongqing Robot Competition
14. 2024 / First Prize, National Business Elite Challenge Finals
15. 2024 / Third Prize, National Intelligent Transportation Innovation and Entrepreneurship Competition
16. 2024 / Third Prize, China Graduate Electronics Design Competition
17. 2024 / Second Prize, University Entrepreneurship Competition
18. 2024 / First Prize, RAICOM Robot Developer Competition
19. 2023 / First and Second Prizes, National Undergraduate Mathematical Contest in Modeling, Chongqing
20. 2023 / Third Prize, China Internet+ College Students' Innovation and Entrepreneurship Competition, Campus Round
21. 2023 / Third Prize, China Software Cup Western Division